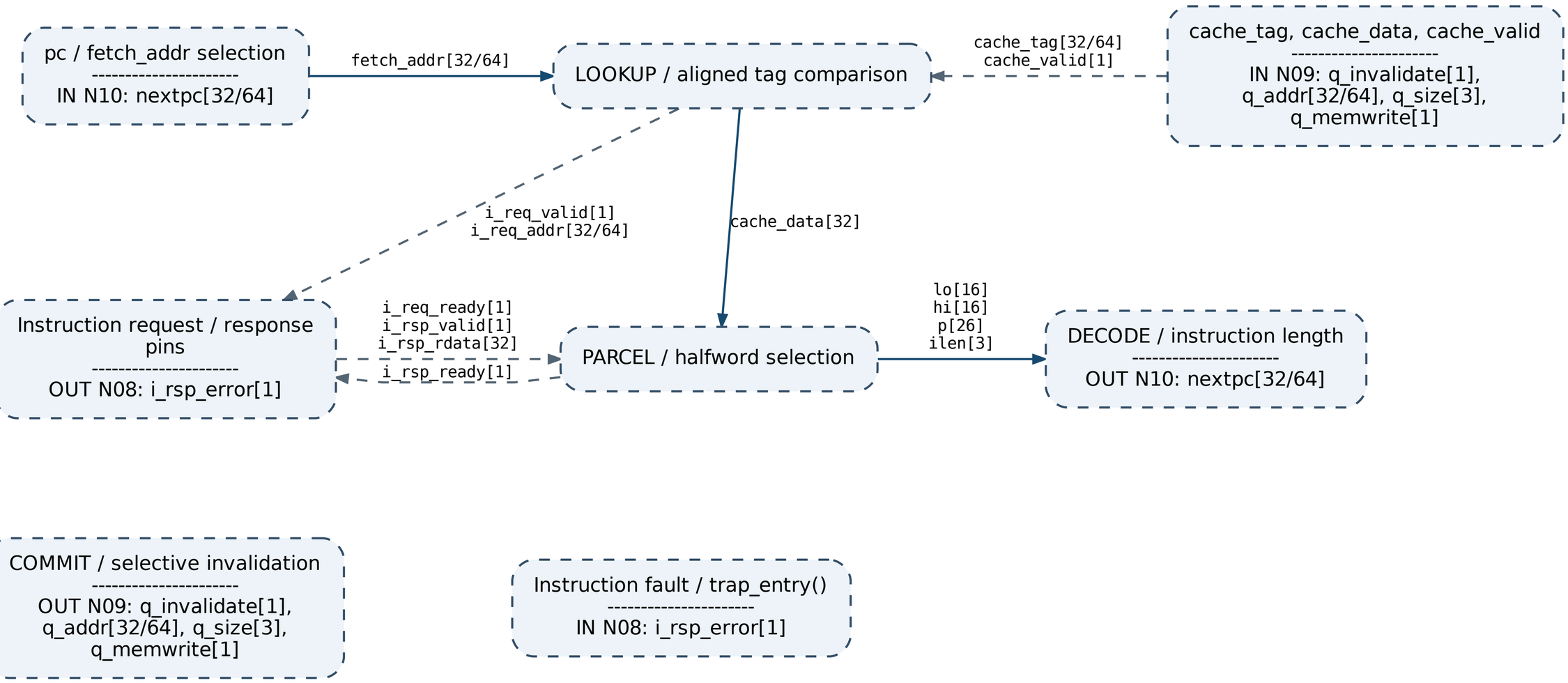
MICA 3.0 development | 02 / 20 | Instruction fetch and parcel assembly

ISA 2.0 compatibility baseline | E32: XLEN=32, XREG/MUL/FLOW/BITOPS/TRAP/REG_PARITY=1 | implemented RTL

RTL source set 5fbe88e7b5ecdbe5 | internal clock: clk_i rising edge; reset: rst_ni active low

Fetch requests are four-byte aligned and return one 32-bit word. One outstanding request.
FAST_FETCH=1: an aligned long instruction captures both parcels without another lookup.
Cross-word long instructions fetch the second word. Redirects use tags; overlapping stores and fence.i invalidate.
Request fields remain stable under backpressure; response errors never retire the instruction.

Blue: data. Dashed gray: control/handshake. Red: fault. Repeated Nxx labels connect without a wire. Crossings do not join.